

pollution.

pollution and control of air, water, and soil pollution.

pollution is an undesirable and excessive addition of substances to water, air and land that adversely alter the natural quality of the environment. The agent that pollute is called pollutant.

Air pollution.

Mainly 2 categories.

- 1) Gaseous. Remain in gaseous state in normal temperature and pressure.

Ex: 1. CO_2

- * released - burning of fossil fuel.
- * blanket around the earth.
- * It stops the heat from escaping and causes temperature to rise called "green house effect".

2. Oxides of sulphur.

- * SO_2 and SO_3 are harmful.
- * react with water and form sulphuric acid $\rightarrow$ cause $\rightarrow$ acid rain
- * harmful to plants fishes etc.

3. Oxides of Nitrogen

- * dilute of nitric acid cause acid rain
- * harmful for human

4 CO

- * colorless, odorless, nonirritant but highly toxic gas.
- * diffuse to the blood stream and combine with Hemoglobin and stops O₂ transport inside the body.

2) particulate.

- substances that occur in the solid or liquid form

Example

Smog.

- Incomplete combustion of petrol causes smoke.
- mainly - suspension of carbon particles.
- Reduces the amount of sunlight reaching the earth.
- combines with water vapor, it causes "smog"
- cause lung disorders in human.

control of air pollution

Before releasing the gasses into atmosphere from various industries

1. separate the pollutants from harmless gasses.
2. convert the pollutants in unharmed products.

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Water pollution

1. compounds - Hg, As, Pb
 - released by the industries as iron and steel industry
 - cause - skin disease and irreversible brain damage.
2. Inorganic Nitrate and phosphate.
 - released by - chemical plants and cause eutrophication
 - condition of sudden increase of nutrients in lakes
 - It affect fauna and flora.
3. Organochlorines
 - released by the excess use of chemical for agriculture
 - pass through food chain
 - Affect nervous system.
4. animal excreta.
 - released with untreated sewage.
 - contains pathogens and toxic wastes.
 - causes epidemic diseases like cholera.
5. oils in sea water.
 - released by oil tankers (being loaded or unloaded)
 - kill sea birds, plants and animals in oceans.
6. Hot water.
 - released - electric power plants.
 - cause - sudden increase in temperature in the water body.
 - affect - fauna and flora.

control of water pollution.

1. Sewage and factory wastes should be treated before flow into water ways.
2. various pollutants should be recycled instead of dumping in to water.